COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The application uses Next.js, TypeScript, Prisma and SQLite. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only, serving a single warehouse and the four roles above; transactional guards — stock checks and one confirmation transaction (§5) — and role gates (FR-07) enforce data integrity.

2. Development process

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Coursework kickoff	Scope, responsibilities and contracts recorded (increment register)
17 Oct 2026 review	Order and stock functions ready for review	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Review found a confirmation endpoint that accepted repeat submits and a missing ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	All four role workflows runnable together	All four role workflows verified in the 11 Nov regression, driver operations included (browser checks)
Driver and privacy, 2 Nov 2026	Driver workflow and data minimisation outstanding	Driver assignment and data-minimisation checks pass (tests)
Documentation hardening, 10–11 Nov 2026	Review found diagrams and traceability diverging from the implementation	Diagrams and traceability aligned (traceability)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date CSV export in three interface interactions (walkthrough record)
CI health tests, 11–12 Nov 2026	The CI health record had no regression test	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

Functional requirements and verification

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	Test — lifecycle, stock and cancellation (cancel tests)
FR-03	number orders and shipments and show owner tracking (the staff member or driver holding each order)	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	Test — inventory, replenishment and alerts
FR-05	record event notifications and allow reading	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	Test — aggregates and export
FR-07	authenticate and enforce role and object access	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	Test — assignment and delivery

- NFR-01 performance — API p95 ≤2 s, first contentful paint ≤3 s. - NFR-02 CI health — green build, test and browser runs recorded per increment (observations). - NFR-03 least privilege and data minimisation — role gates on every mutating endpoint; staff responses omit customer identity and delivery address (privacy tests). - NFR-04 interaction budget — at most three interface interactions per core operation (walkthrough, §8). - NFR-05 process quality — CI-verified increments (workflow).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	scheduling plan	increment register
RK-02 contract rework	Transition tests pin FR-02	transition tests
RK-03 test-week capacity	Traceability	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	Seed data regression-tested	seed regression test
RK-06 coursework assignments and test-week squeeze	Weekly re-planning	risk plan
RK-07 stack unfamiliarity	Setup guide and CI lower ramp-up cost	setup guide
RK-08 report and video compete for time	Risk plan	report and video plan

4. System models

The four layers — Role-specific pages, HTTP routes and authentication, Order, inventory, driver, notification and reporting business handlers, Prisma/SQLite persistence — separate interface, application and data concerns. Figures 1–5 answer, in order: role functions; transitions and stock effects; data relationships; the confirmation transaction; layer connections.

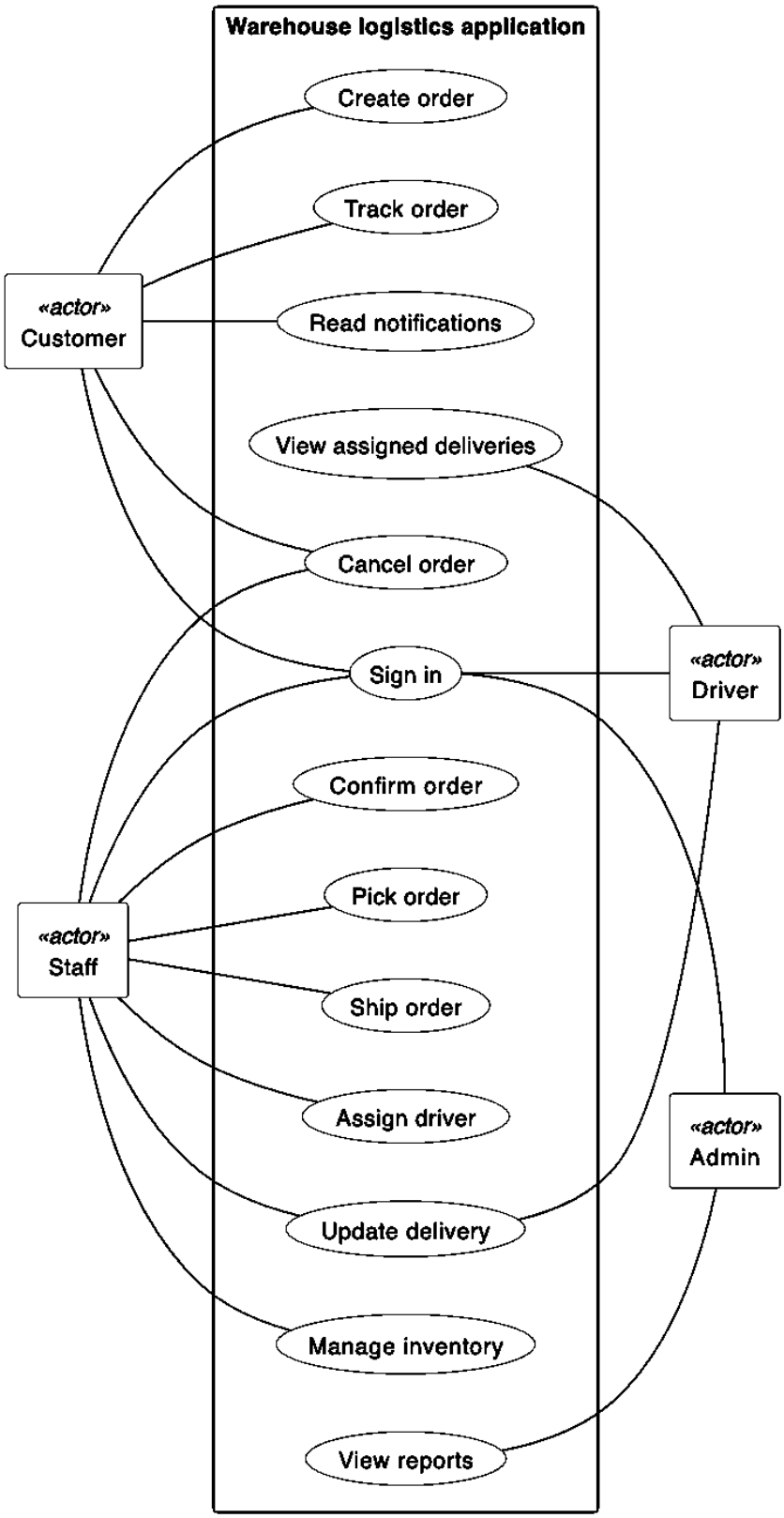


Figure 1. Role functions. Source.

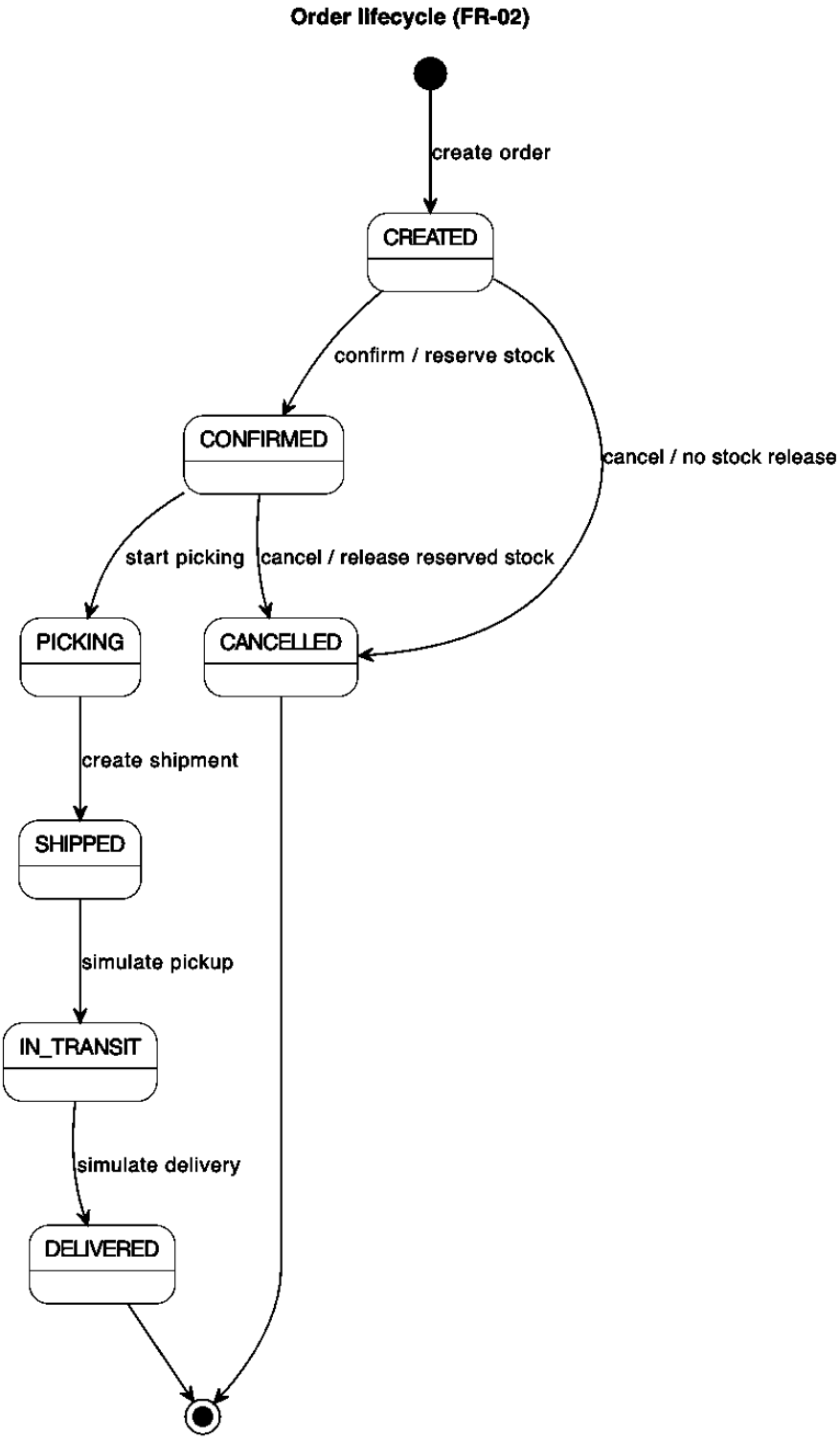


Figure 2. Order transitions and stock effects. Source.

Each item references one product; each order requires at least one item; orders have 0..1 shipment; products have 0..* ledger entries; users receive notifications, and each notification relates to at most one order.

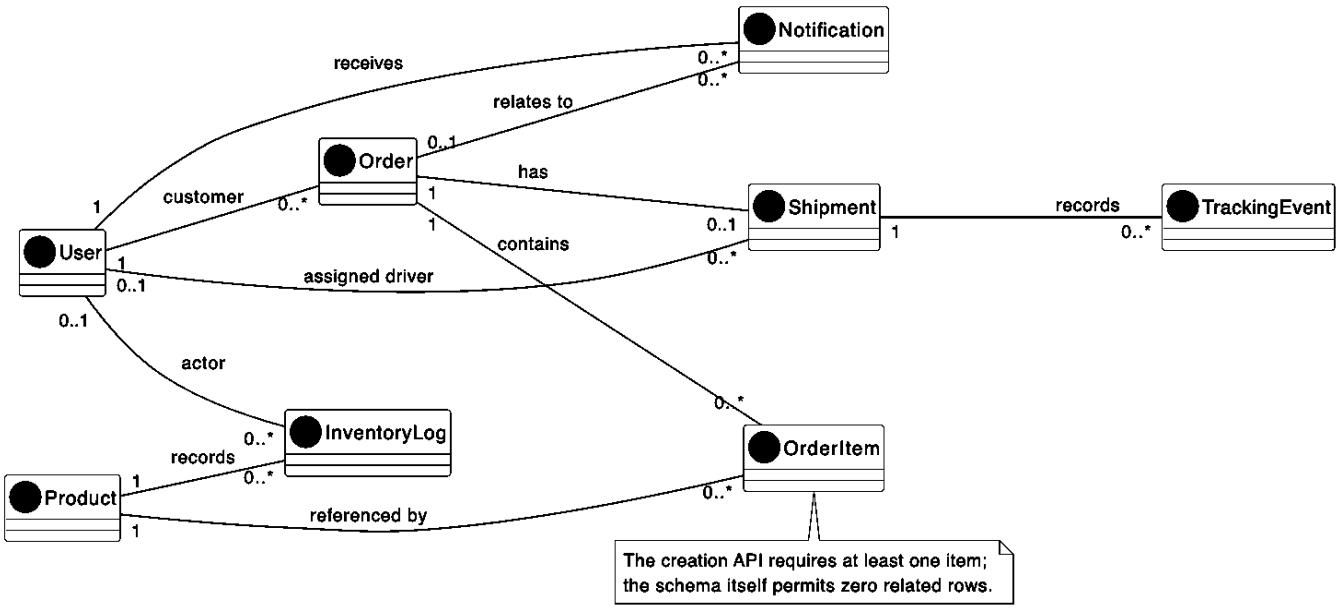


Figure 3. Data relationships. Source.

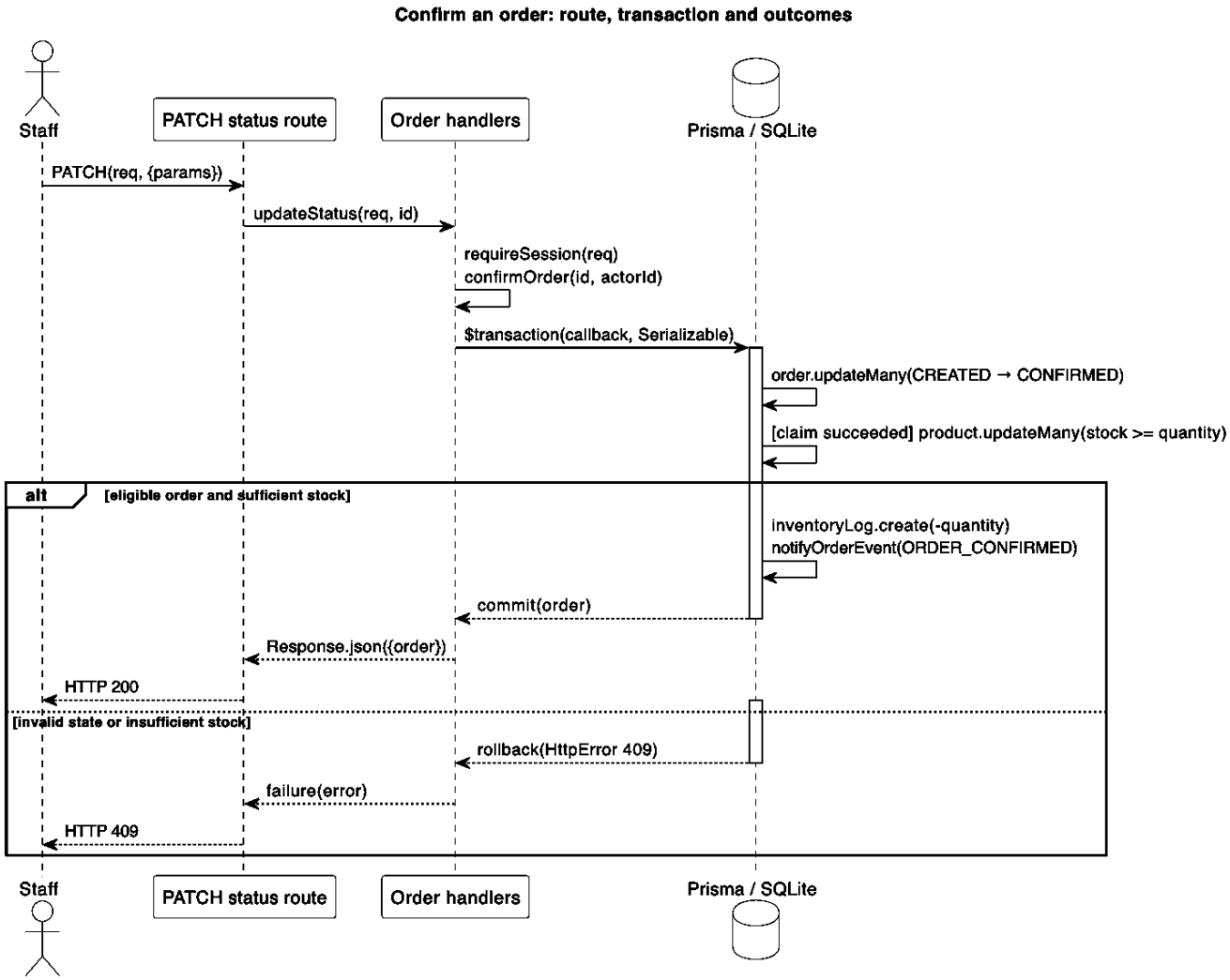


Figure 4. Confirmation transaction. Source.

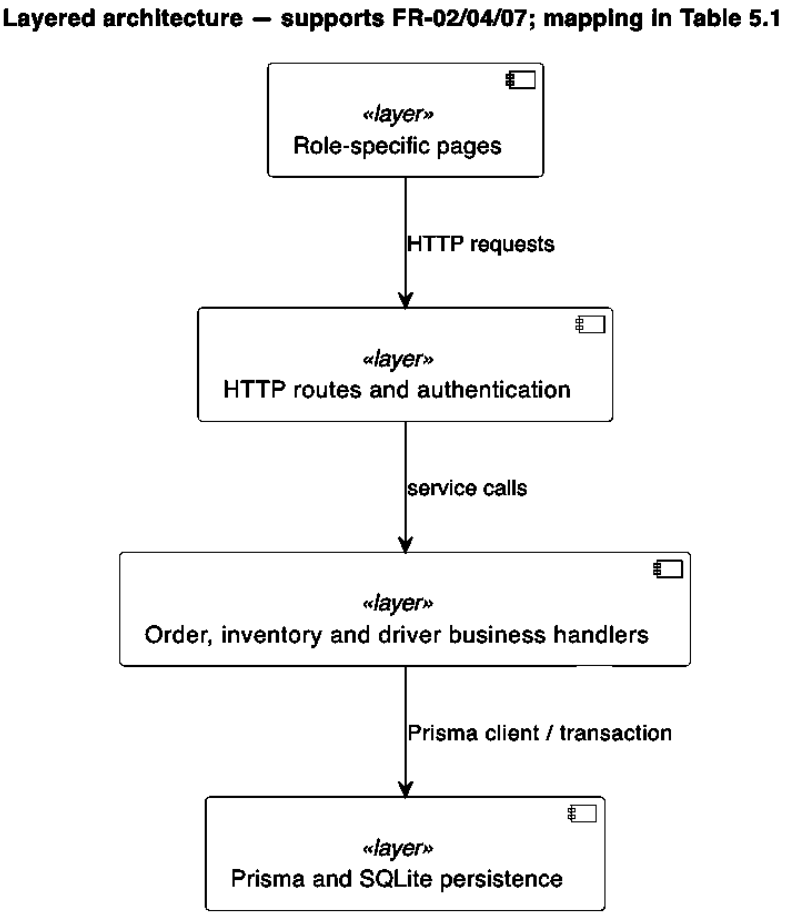


Figure 5. Four layers. Source.

5. Implementation

Stock is checked at creation and reserved at confirmation; the confirmation transaction (order-api.ts:245–293) enforces both. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED, with CANCELLED reachable from CREATED or CONFIRMED; cancellation per the matrix below. Shipments are created when staff ship.

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: single transaction writing state, stock, ledger and notifications; rolls back on insufficient stock; repeated confirmation is rejected by the guards (409)	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 ledger	order-api.ts:281, 398	Inventory

6. Verification

Verification checks requirements against the implementation; validation exercises the workflows through the interface (§8). CI records 38 unit and 140 integration checks; 2 delivery checks verify packaging and evidence integrity (delivery checks); the regression log reads "Tests 38 passed (38)" and the walkthrough record ends "Browser acceptance passed: 26 checks".

In a local-loopback measurement (11 Nov) — 30 sequential requests per endpoint after warm-up, one Chromium instance — API p95 was 3.2–13.6 ms across endpoints and first-contentful-paint p95 was 64–96 ms, against local-deployment targets of at most 2 s and 3 s (§9). NFR-02 is evidenced by recorded green build, test and browser runs (observations).

7. Evolution

The 17 Oct 2026 review exercised the running app and found a confirmation endpoint that accepted repeat submits and a missing ledger; the confirmation transaction and retest followed (26–28 Oct). The 10–11 Nov documentation review corrected the diagrams and traceability.

8. Validation

The automated walkthrough exercises eleven core operations — create order; order details; confirm; pick; ship; assign; pickup; delivery; report filter; export report; selected-date export — each by its owning role, at most three in-app interface interactions each (login, navigation, waits and assertions excluded). Its 26 Playwright browser checks cover the eleven operations at one to three checks each, with error and interface checks alongside; expected outcomes are asserted in the script. Results are in the walkthrough record.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagram bases diverged from the schema	Models corrected	10–11 Nov documentation review	Diagran
Stock effects were recorded under FR-04	Moved to FR-02	17 Oct review	Traceal
Trigger and outcome detail was thin	Rows rewritten with concrete triggers and outcomes	10–11 Nov documentation review	Registe
Export took four scripted interactions	Simplified to three in-app interactions; issue closed	11 Nov automated walkthrough	Issue #
The CI health record had no regression test	Suite reached 38	11 Nov regression work	Test his
NFR criteria assumed a hosted deployment model	Reset to local-deployment budgets (§6)	10–11 Nov documentation review	Require

10. Evidence

Traceability · Baseline record — the recorded 12 Nov suite: 38 unit, 140 integration, 2 delivery and 26 browser checks.